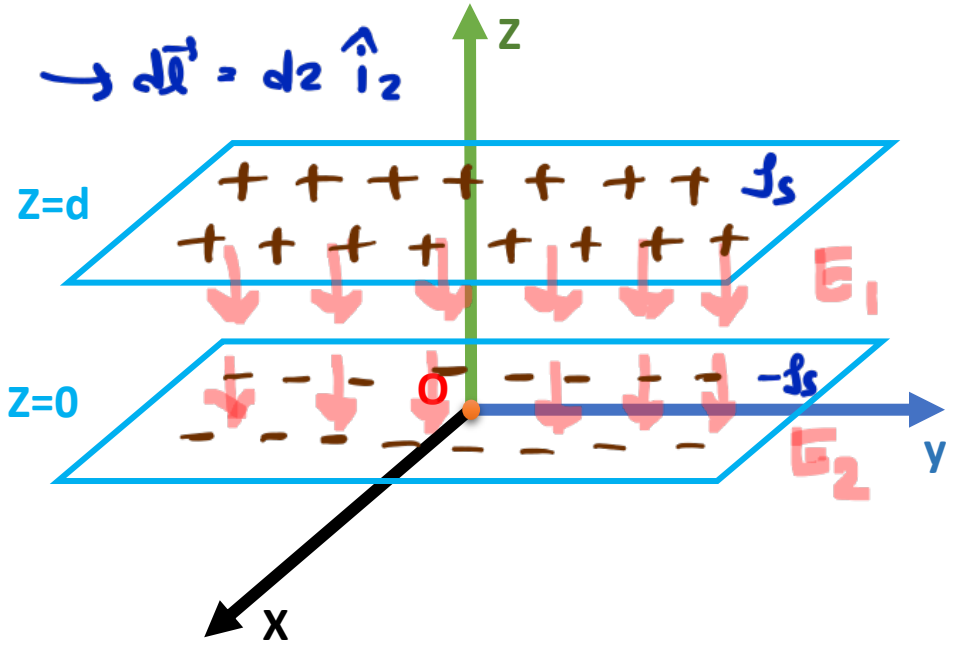
Parallel Plate Capacitor

Outlines

- Structure of Parallel Plate Capacitor
- Capacitance of Parallel Plate Capacitor
- Energy Stored in Parallel Plate Capacitor
- Energy Density in Parallel Plate Capacitor

Engineering Funda

Parallel Plate capacitor



$$\sim \vec{E} = \frac{\rho_s}{2\varepsilon_0}(-\hat{i}_z) + \frac{\rho_s}{2\varepsilon_0}(-\hat{i}_z)$$

$$= -\frac{\rho_s}{\varepsilon_0}\hat{i}_z$$

$\sim$ Capacitance C

$$\Rightarrow C = Q/V \quad \text{— (A)}$$

$\sim$ Charge Q

$$\Rightarrow Q = \rho_s A \quad \text{— (2)}$$

$\sim$ voltage V

$$\Rightarrow V = -\int \vec{E}\cdot d\vec{l}$$

$$= -\int_0^d -\frac{\rho_s}{\varepsilon_0}\hat{i}_z \cdot dz\,\hat{i}_z$$

$$= \frac{\rho_s}{\varepsilon_0}[z]_0^d$$

$$= \frac{\rho_s d}{\varepsilon_0} \quad \text{— (1)}$$

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⇝ Substitute eq.^{n} (1) & (2) in eq.^{n} (A)

$$\Rightarrow C = \frac{Q}{V} = \frac{\cancel{\rho_s} A}{\frac{\cancel{\rho_s} d}{\varepsilon_0}} = \varepsilon_0 \frac{A}{d}.$$

⇝ For dielectric material ε_r

$$\boxed{\Rightarrow C = \varepsilon_0 \varepsilon_r \frac{A}{d}} = \frac{Q}{V}$$

⇝ Energy stored in Capacitor E

$$\Rightarrow E = \frac{1}{2} C V^2 = \frac{1}{2} \frac{Q^2}{C} \quad (J)$$

⇝ Energy density in Capacitor

$$\Rightarrow \omega = \frac{1}{2} \varepsilon_0 |E|^2 \quad (J/m^3)$$

Engineering Funda

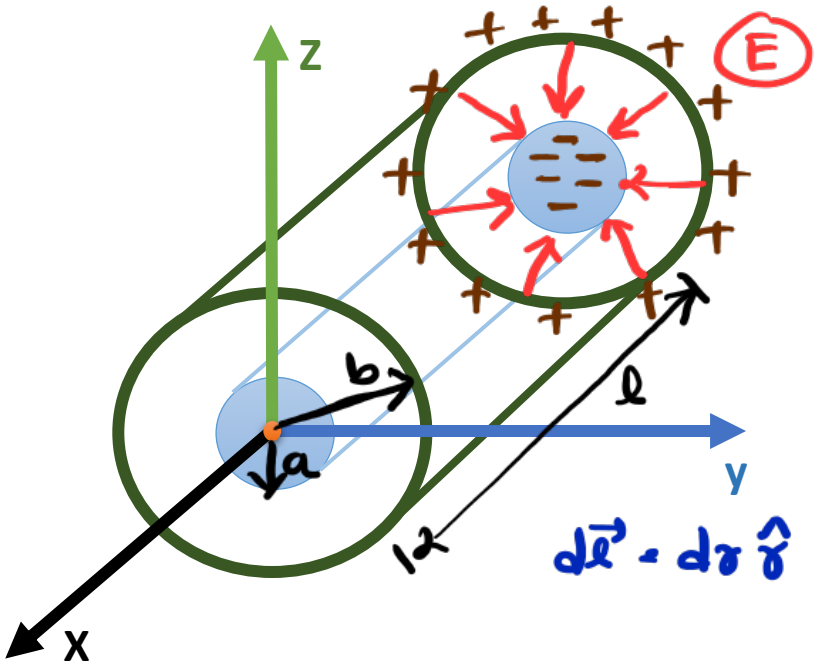
Coaxial Cable Capacitor

Outlines

- Structure of Coaxial Cable Capacitor
- Capacitance of Coaxial Cable Capacitor
- Solved Example of Coaxial Cable Capacitor

Engineering Funda

Coaxial Cable capacitor



$d\vec{l} = dr\,\hat{r}$

⇝ Capacitance C

$\Rightarrow C = Q/V$ — (A)

⇝ Charge Q

$\Rightarrow Q = \rho_L l$ — (1)

⇝ Voltage V

$\Rightarrow V = -\int \vec{E}\cdot d\vec{l}$

⇝ Electric field due to line charge ρ_L

$$\Rightarrow \vec{E} = -\frac{\rho_L}{2\pi\epsilon_0 r}\,\hat{r}$$

⇝ Hence, voltage V is given by :

Engineering Funda

$$\Rightarrow V = -\int_a^b \frac{-\rho_L}{2\pi\varepsilon_0 r}\,\hat{r}\cdot dr\,\hat{r}$$

$$= \frac{\rho_L}{2\pi\varepsilon_0}\int_a^b \frac{dr}{r}$$

$$= \frac{\rho_L}{2\pi\varepsilon_0}\left[\ln r\right]_a^b$$

$$= \frac{\rho_L}{2\pi\varepsilon_0}\left[\ln b - \ln a\right]$$

$$= \frac{\rho_L}{2\pi\varepsilon_0}\left[\ln(b/a)\right] \quad \text{— (2)}$$

$\rightsquigarrow$ Put eqn (1) and (2) in eqn (A)

$$\Rightarrow C = \frac{Q}{V} = \frac{\cancel{\rho_L} l}{\frac{\cancel{\rho_L}}{2\pi\varepsilon_0}\ln(b/a)} = \frac{2\pi\varepsilon_0 l}{\ln(b/a)} = \frac{2\pi\varepsilon_0\varepsilon_r l}{\ln(b/a)}$$

Engineering Funda

Question – If the inner radius of the coaxial cable is 1mm and the outer radius is 3mm for coaxial cable with dielectric material 4.5. Then calculate capacitance per unit length.

$\rightsquigarrow$ $a = 1\,mm = 10^{-3}\,m$

$b = 3\,mm = 3 \times 10^{-3}\,m$

$\varepsilon_r = 4.5$

$\frac{C}{l} = ?$

$$\rightsquigarrow \frac{C}{l} = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(b/a)} = \frac{2\pi \times 8.854 \times 10^{-12} \times 4.5}{\ln(3)} = 0.27 \times 10^{-9}\,F = 0.27\,nF$$

Engineering Funda

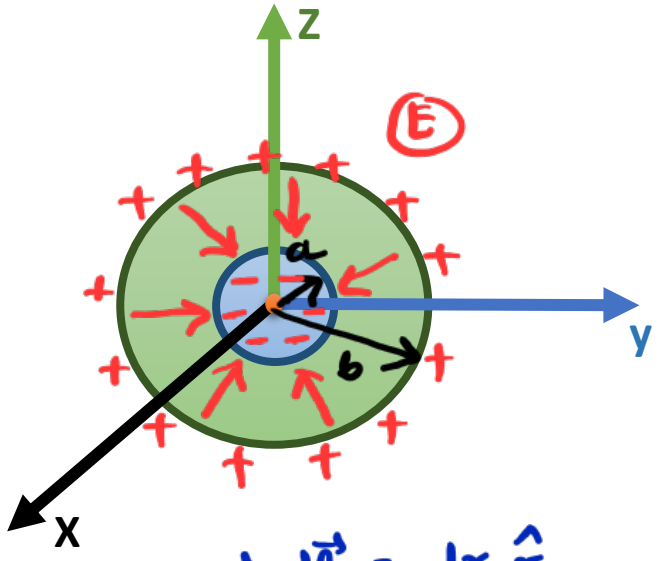
Spherical Capacitor

Outlines

- Structure of Spherical Capacitor
- Capacitance of Spherical Capacitor
- Capacitance due to Multiple Spheres
- Solved Example due to Spherical Capacitor

Engineering Funda

Spherical capacitor



$\rightarrow d\vec{l} = dr\,\hat{r}$

⇝ Capacitance C

$$\Rightarrow C = \frac{Q}{V}$$

⇝ Electric field due to Spherical shell is given by:

$$\Rightarrow \vec{E} = \frac{-1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r}$$

⇝ Voltage V

$$\Rightarrow V = -\int \vec{E}\cdot d\vec{l}$$

$$= -\int_a^b \frac{-1}{4\pi\varepsilon_0}\frac{Q}{r^2}\hat{r}\cdot dr\,\hat{r}$$

$$= \frac{Q}{4\pi\varepsilon_0}\int_a^b \frac{dr}{r^2}$$

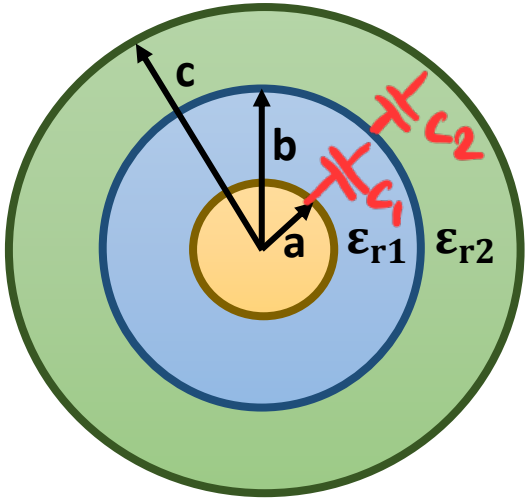
$$= \frac{Q}{4\pi\varepsilon_0}\left[\frac{-1}{r}\right]_a^b = \frac{Q}{4\pi\varepsilon_0}\left[\frac{1}{a}-\frac{1}{b}\right]$$

⇝ Hence, Capacitance C

$$\Rightarrow C = \frac{Q}{V} = \frac{4\pi\varepsilon_0}{\left[\frac{1}{a}-\frac{1}{b}\right]}$$

EngineeringFunda

Capacitance due to Multiple Spheres



$\rightsquigarrow$ C_1 is Capacitance betn a to b.

$$\Rightarrow C_1 = \frac{4\pi\varepsilon_0\varepsilon_{r1}}{\left[\frac{1}{a}-\frac{1}{b}\right]}$$

$\rightsquigarrow$ C_2 is Capacitance betn b to c

$$\Rightarrow C_2 = \frac{4\pi\varepsilon_0\varepsilon_{r2}}{\left[\frac{1}{b}-\frac{1}{c}\right]}$$

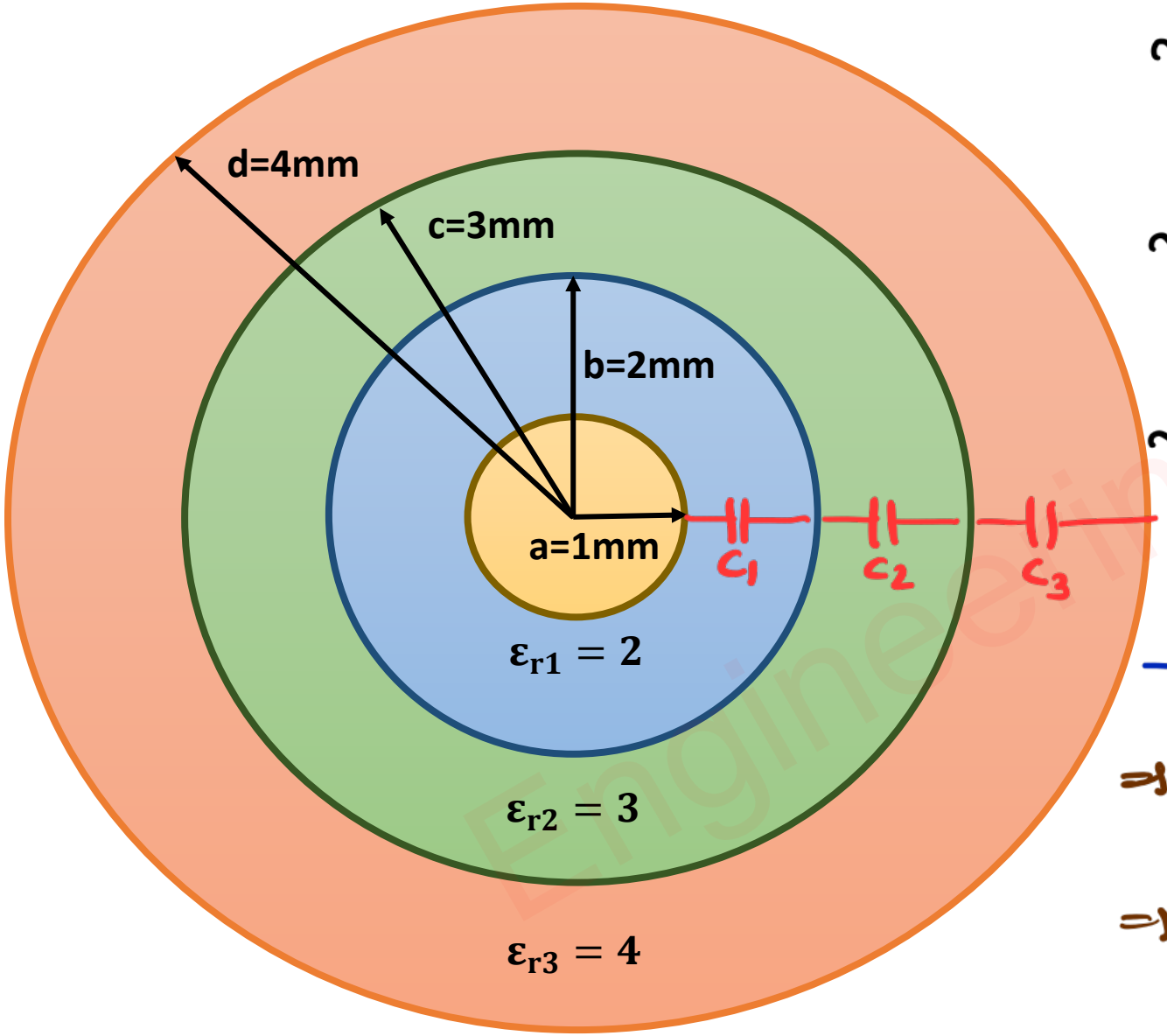
$\rightsquigarrow$ For Series Connection C

$$\Rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow \frac{1}{C} = \frac{\left[1/a - 1/b\right]}{4\pi\varepsilon_0\varepsilon_{r1}} + \frac{\left[\frac{1}{b}-\frac{1}{c}\right]}{4\pi\varepsilon_0\varepsilon_{r2}}$$

Engineering Funda

Question – Find the capacitor for given structure.



$$\rightsquigarrow C_1 = \frac{4\pi\varepsilon_0\varepsilon_{r1}}{\left[\frac{1}{a}-\frac{1}{b}\right]} = \frac{4\pi\varepsilon_0(2)}{\left[\frac{1}{1}-\frac{1}{2}\right]}\times10^{-3} = 16\pi\varepsilon_0\times10^{-3}$$

$$\rightsquigarrow C_2 = \frac{4\pi\varepsilon_0\varepsilon_{r2}}{\left[\frac{1}{b}-\frac{1}{c}\right]} = \frac{4\pi\varepsilon_0(3)}{\left[\frac{1}{2}-\frac{1}{3}\right]}\times10^{-3} = 72\pi\varepsilon_0\times10^{-3}$$

$$\rightsquigarrow C_3 = \frac{4\pi\varepsilon_0\varepsilon_{r3}}{\left[\frac{1}{c}-\frac{1}{d}\right]} = \frac{4\pi\varepsilon_0(4)}{\left[\frac{1}{3}-\frac{1}{4}\right]}\times10^{-3} = 192\pi\varepsilon_0\times10^{-3}$$

→ Series Connection

$$\Rightarrow \frac{1}{C} = \frac{1}{C_1}+\frac{1}{C_2}+\frac{1}{C_3} = \frac{1}{\pi\varepsilon_0\times10^{-3}}\left[\frac{1}{16}+\frac{1}{72}+\frac{1}{192}\right]$$

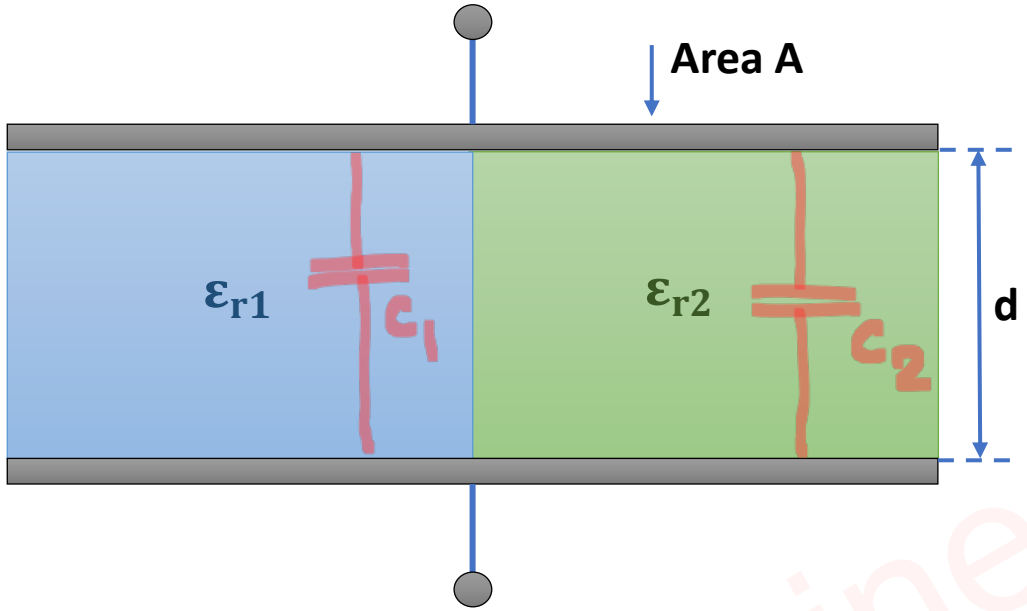
$$\Rightarrow C = \frac{\pi\varepsilon_0\times10^{-3}}{\left[\frac{1}{16}+\frac{1}{72}+\frac{1}{192}\right]} = 12.25\times3.14\times8.854\times10^{-12}\times10^{-3}$$

$$\Rightarrow C = 0.34\times10^{-12}\,F = 0.34\,pF$$

Engineering Funda

Examples of Capacitor & Capacitance

Question 1 – Find the Equivalent Capacitance between Two terminals.

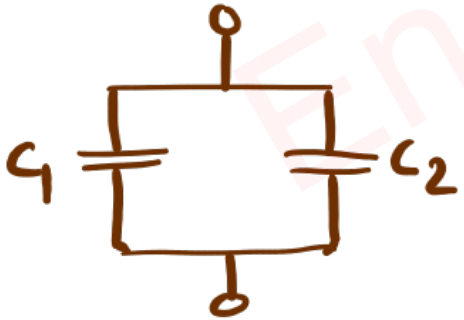


$$\rightarrow C_1 = \varepsilon_0 \varepsilon_{r1} \left(\frac{A/2}{d}\right) = \varepsilon_0 \varepsilon_{r1} \left(\frac{A}{2d}\right)$$

$$\rightarrow C_2 = \varepsilon_0 \varepsilon_{r2} \left(\frac{A/2}{d}\right) = \varepsilon_0 \varepsilon_{r2} \left(\frac{A}{2d}\right)$$

$$\rightarrow C = C_1 + C_2$$

$$= \varepsilon_0 \left(\frac{A}{2d}\right) (\varepsilon_{r1} + \varepsilon_{r2})$$



$$\Rightarrow C = C_1 + C_2$$

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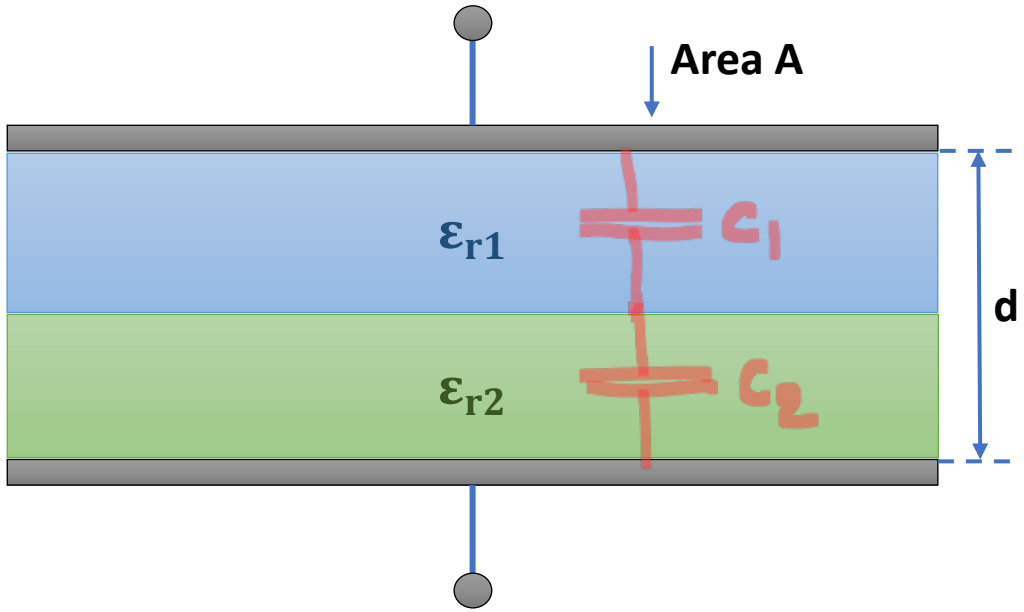
Question 2 – Find the Equivalent Capacitance between Two terminals.



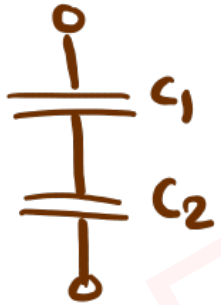
$$\rightsquigarrow C_1 = \varepsilon_0 \varepsilon_{r1} \left(\frac{A}{d/2}\right) = \varepsilon_0 \varepsilon_{r1} \left(\frac{2A}{d}\right)$$

$$\rightsquigarrow C_2 = \varepsilon_0 \varepsilon_{r2} \left(\frac{A}{d/2}\right) = \varepsilon_0 \varepsilon_{r2} \left(\frac{2A}{d}\right)$$

$$\rightsquigarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

$$\Rightarrow \frac{1}{C} = \frac{1}{\varepsilon_0\left(\frac{2A}{d}\right)} \left[\frac{1}{\varepsilon_{r1}} + \frac{1}{\varepsilon_{r2}}\right]$$

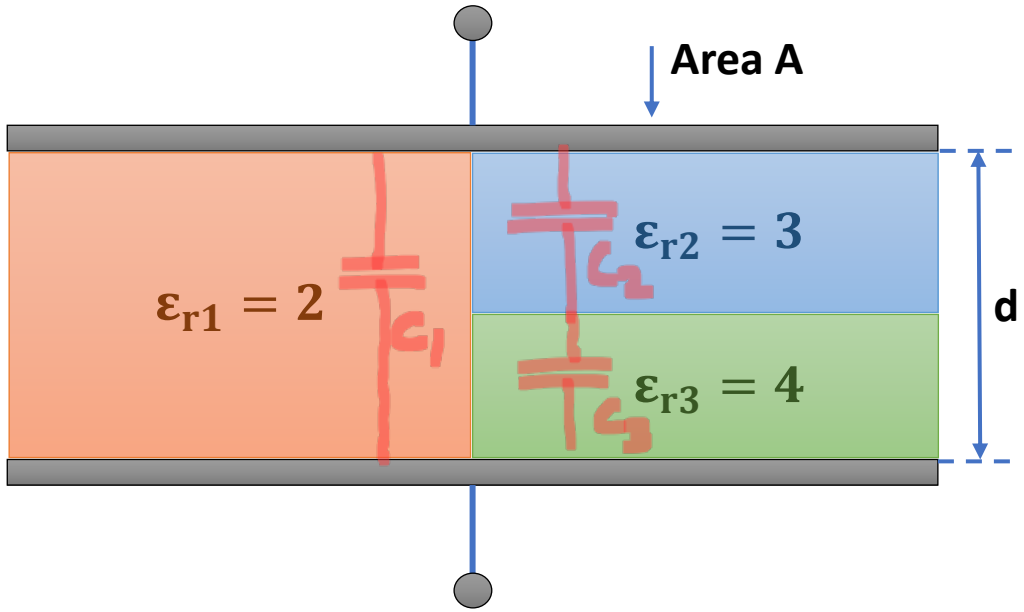
$$\Rightarrow C = \varepsilon_0\left(\frac{2A}{d}\right) \left[\frac{\varepsilon_{r1}\varepsilon_{r2}}{\varepsilon_{r1} + \varepsilon_{r2}}\right]$$



$$\rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$$

Engineering Funda

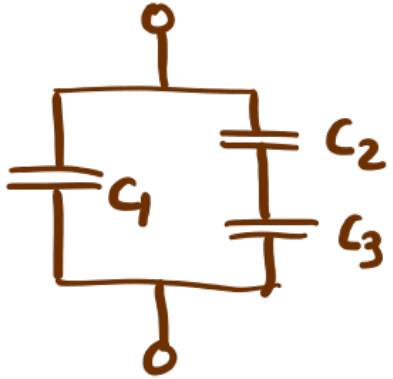
Question 3 – Find the resultant dielectric constant with given capacitor.



$$\rightarrow C_1 = \varepsilon_0 \varepsilon_{r1} \left(\frac{A/2}{d}\right) = \varepsilon_0 \left(\frac{A}{d}\right)$$

$$\rightarrow C_2 = \varepsilon_0 \varepsilon_{r2} \left(\frac{A/2}{d/2}\right) = 3\left(\varepsilon_0 (A/d)\right)$$

$$\rightarrow C_3 = \varepsilon_0 \varepsilon_{r3} \left(\frac{A/2}{d/2}\right) = 4\left(\varepsilon_0 (A/d)\right)$$



$$\Rightarrow C = C_1 + \frac{C_2 C_3}{C_2 + C_3}$$

$$= \varepsilon_0 \varepsilon_r \frac{A}{d}$$

$$\rightarrow C = \varepsilon_0 (A/d) \left[1 + \frac{3 \times 4}{3+4}\right]$$

$$= \varepsilon_0 (A/d) [1 + 1.71]$$

$$= 2.71 \left[\varepsilon_0 (A/d)\right]$$

$$\rightsquigarrow \boxed{\varepsilon_r = 2.71}$$

Engineering Funda

Examples of Capacitor & Capacitance

Question 4 – A Dielectric Slab with $500\text{mm} \times 500\text{mm}$ Cross section is 0.4m long. The Slab is subjected to a uniform electric field of $\vec{E} = 6\widehat{a_x} + 8\widehat{a_y}$ kV/mm. The relative permittivity of the dielectric material is 2. The energy stored in dielectric is ....88.54 J

$$\rightsquigarrow \omega = \frac{1}{2}\varepsilon_0\varepsilon_r |E|^2$$

$$= \frac{1}{\cancel{2}} \times 8.854\times10^{-12} \times \cancel{2} \times 10^{14}$$

$$= 885.4 \ J/m^3$$

$$|\vec{E}| = \sqrt{6^2 + 8^2}$$

$$= 10 \times \frac{10^3}{10^{-3}} \ \frac{V}{m}$$

$$= 10^7 \ V/m$$

$$\rightsquigarrow E = \int \omega \, dV = \omega V$$

$$= 885.4 \times 0.5 \times 0.5 \times 0.4$$

$$= 88.54 \ J$$

Engineering Funda

Question 5 – C_0 is capacitance of parallel plate with air as dielectric as shown in figure a, if half of the region is filled with ε_r as shown in figure b. The modified expression of capacitance is $C = kC_0$, where k is ....$\frac{1}{2}(1+\varepsilon_r)$....

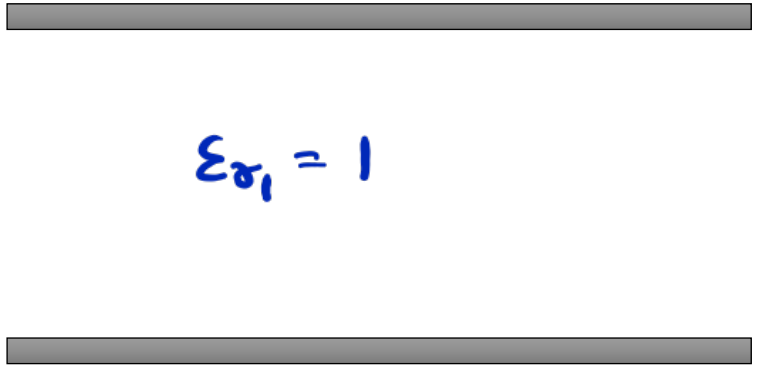


Figure a

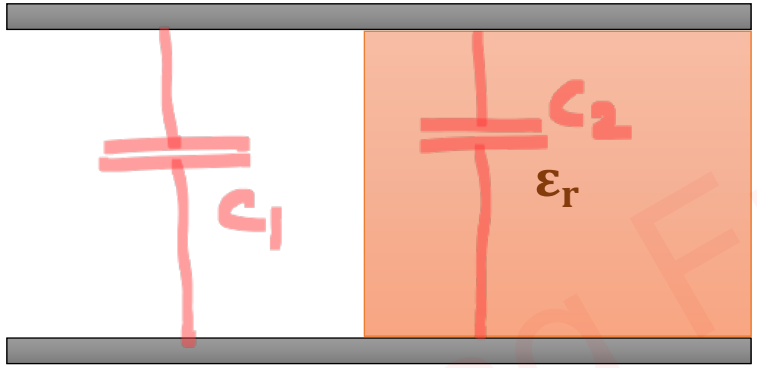


Figure b

$\rightarrow C_0 = \varepsilon_0 \varepsilon_{r_1} A/d$

$= \varepsilon_0 \frac{A}{d}$

$\rightarrow C = C_1 + C_2$

$= \varepsilon_0 \left(\frac{A/2}{d}\right) + \varepsilon_0 \varepsilon_r \left(\frac{A/2}{d}\right)$

$= \varepsilon_0 \left(\frac{A}{d}\right) \left(\frac{1}{2} + \frac{\varepsilon_r}{2}\right)$

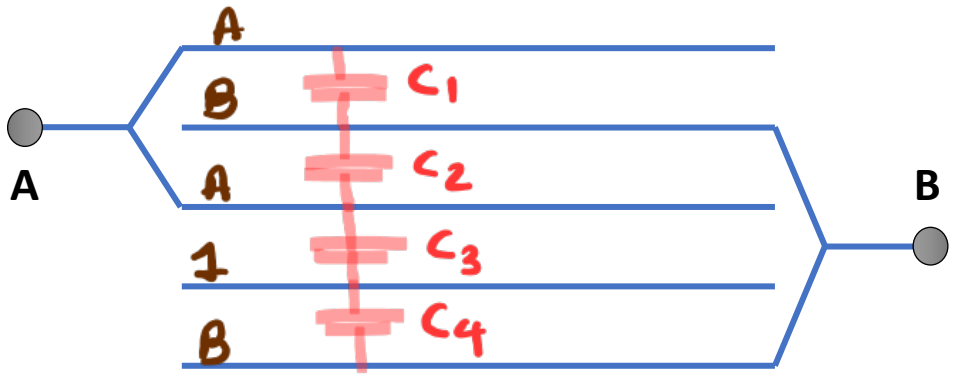
$= C_0 \left(\frac{1}{2} + \frac{\varepsilon_r}{2}\right)$

$\rightarrow k = \frac{1}{2} + \frac{\varepsilon_r}{2}$

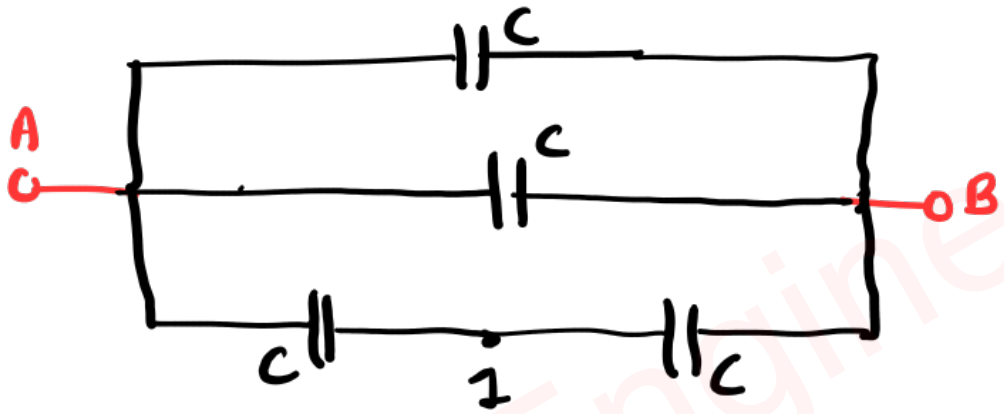
$= \frac{1}{2}(1+\varepsilon_r)$

Engineering Funda

Question 6 – If the spacing between plates is d and area A, find equivalent capacitance between terminal A and B.



$\rightarrow C_1 = C_2 = C_3 = C_4 = \varepsilon_0\left(\frac{A}{d}\right) = C$

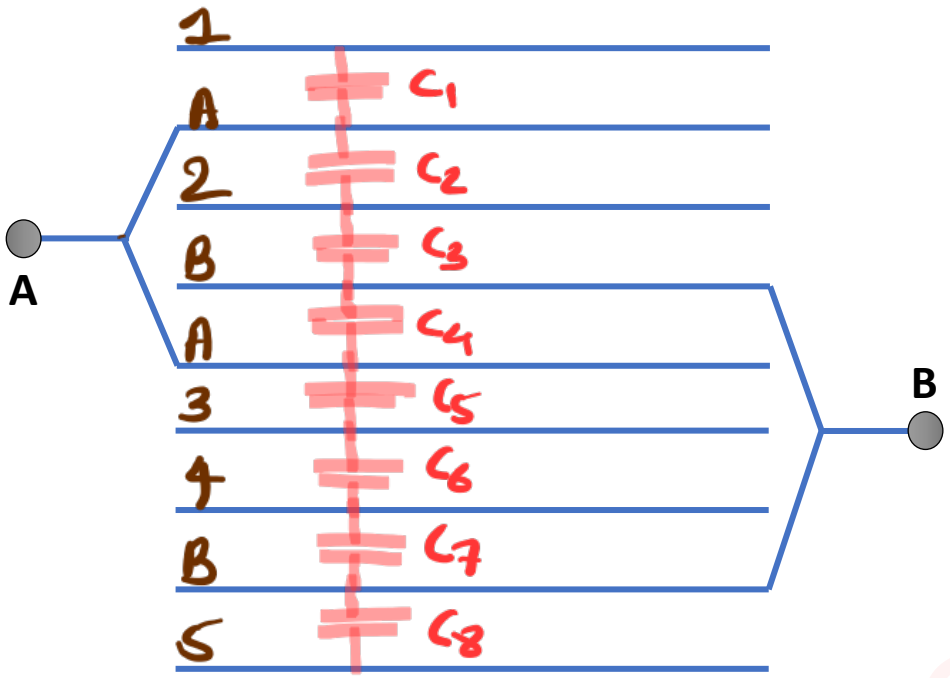


$\rightarrow C_{eq} = C + C + C \approx C$

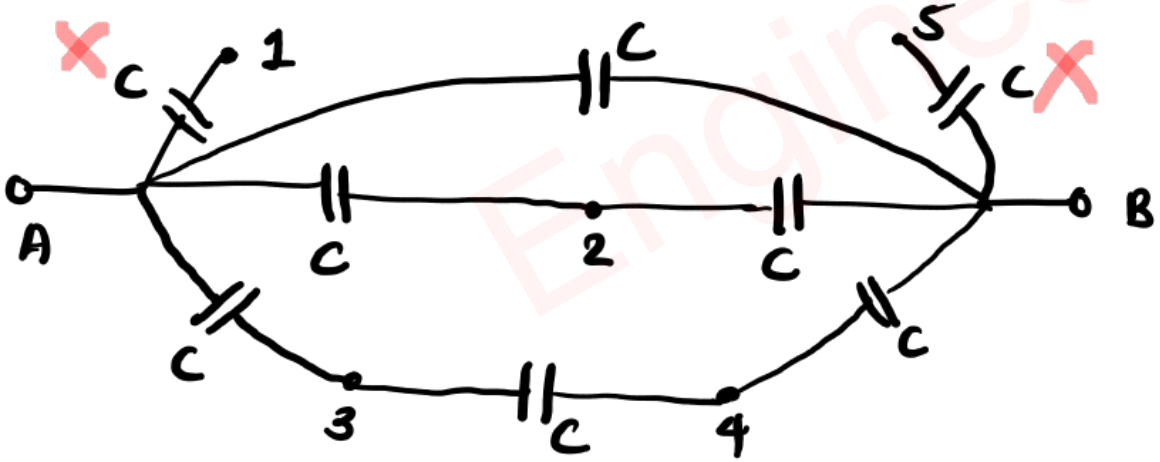
$= \frac{5}{2}C = \frac{5}{2}\varepsilon_0(A/d)$

Engineering Funda

Question 7 – If the spacing between plates is d and area A, find equivalent capacitance between terminal A and B.



$$\rightarrow C_1 = C_2 = C_3 = C_4 = C_5 = C_6 = C_7 = C_8 = \varepsilon_0 \frac{A}{d} = C$$



$$\rightarrow C_{eq} = C + C \approx C + C \approx C \approx C$$

$$= C + \frac{C}{2} + \frac{C}{3}$$

$$= \frac{11}{6} C$$

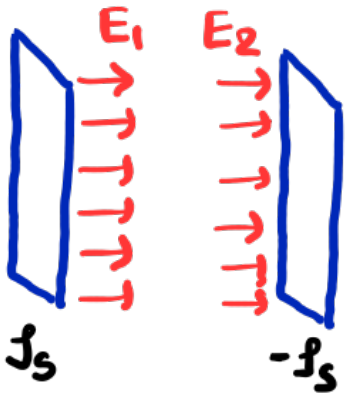
$$= \frac{11}{6} \varepsilon_0 \left(\frac{A}{d}\right)$$

Engineering Funda

Examples of Capacitor & Capacitance

Question 8 – A capacitor is made with a polymeric dielectric having $\varepsilon_r = 2.26$ and dielectric breakdown strength of 50 kV/cm. Rectangular plates have width of 20cm and length of 40cm, The maximum charge in capacitor will be .....

a. 2μC
b. 4μC
c. ✓ 8μC
d. 10μC



$$\rightarrow E = E_1 + E_2$$

$$= \frac{\rho_s}{2\varepsilon_0\varepsilon_r} + \frac{\rho_s}{2\varepsilon_0\varepsilon_r}$$

$$= \frac{\rho_s}{\varepsilon_0\varepsilon_r}$$

$$\Rightarrow \rho_s = \varepsilon_0\varepsilon_r E$$

$$\Rightarrow \frac{Q}{A} = \varepsilon_0\varepsilon_r E$$

$$\Rightarrow Q = \varepsilon_0\varepsilon_r E A$$

$$= 8.854\times10^{-12} \times 2.26 \times 50\times\frac{10^3}{10^{-2}} \times 20\times10^{-2} \times 40\times10^{-2}$$

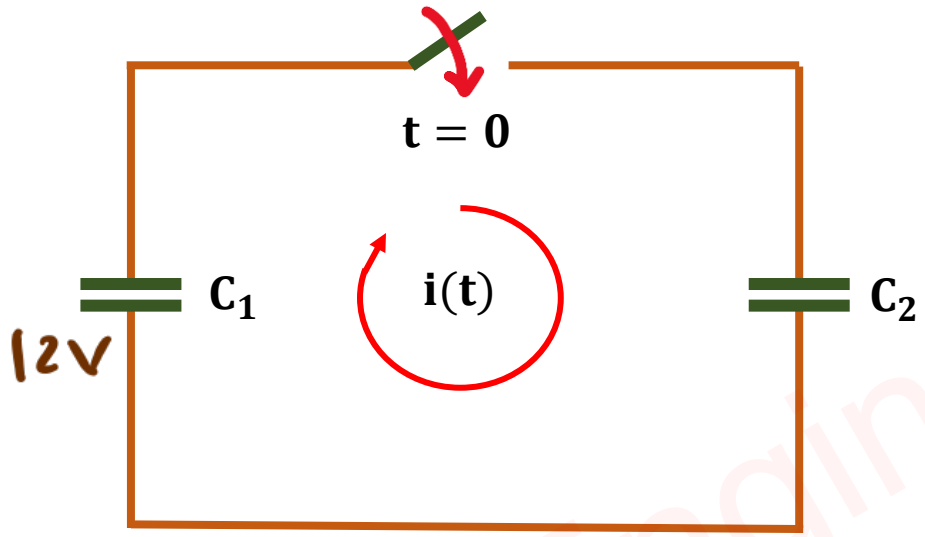
$$= 8\times10^{-6}\,C$$

$$= 8\,\mu C$$

Engineering Funda

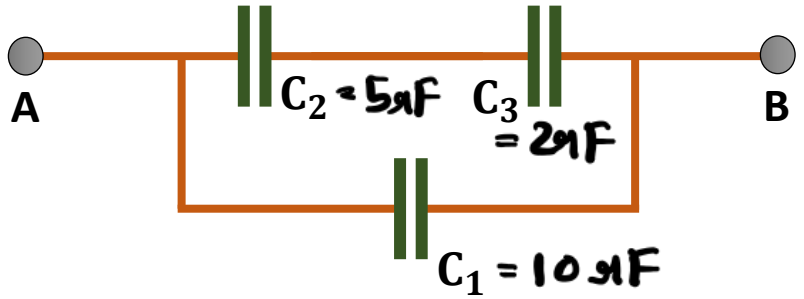
Question 9 – In the following figure, C_1 has been charged to 12V before the ideal switch S is closed at t = 0. The current i(t) is .....

a. Zero
b. A Step Function
c. An Exponentially decaying function [R is available]
d. An Impulse Function



Engineering Funda

Question 10 – Three Capacitor C_1, C_2, and C_3 have values of 10μF, 5μF, and 2μF, respectively. And have breakdown voltages of 10V, 5V, and 2V, Respectively. For the connection shown in the figure, the maximum safe voltage and corresponding charge across two terminals is ……



$\rightarrow C = \frac{Q}{V}$

$\rightsquigarrow V_{1max} = 10V$

$V_{2max} = 5V$

$V_{3max} = 2V$

$\rightsquigarrow Q_{1max} = C_1 V_{1max} = 100\mu C$

$Q_{2max} = C_2 V_{2max} = 25\mu C$

$Q_{3max} = C_3 V_{3max} = 4\mu C$

$Q = Q_1 + Q'$

$= C_1 V_{AB} + 4\mu C$

$= 28\mu C + 4\mu C$

$= 32\mu C$

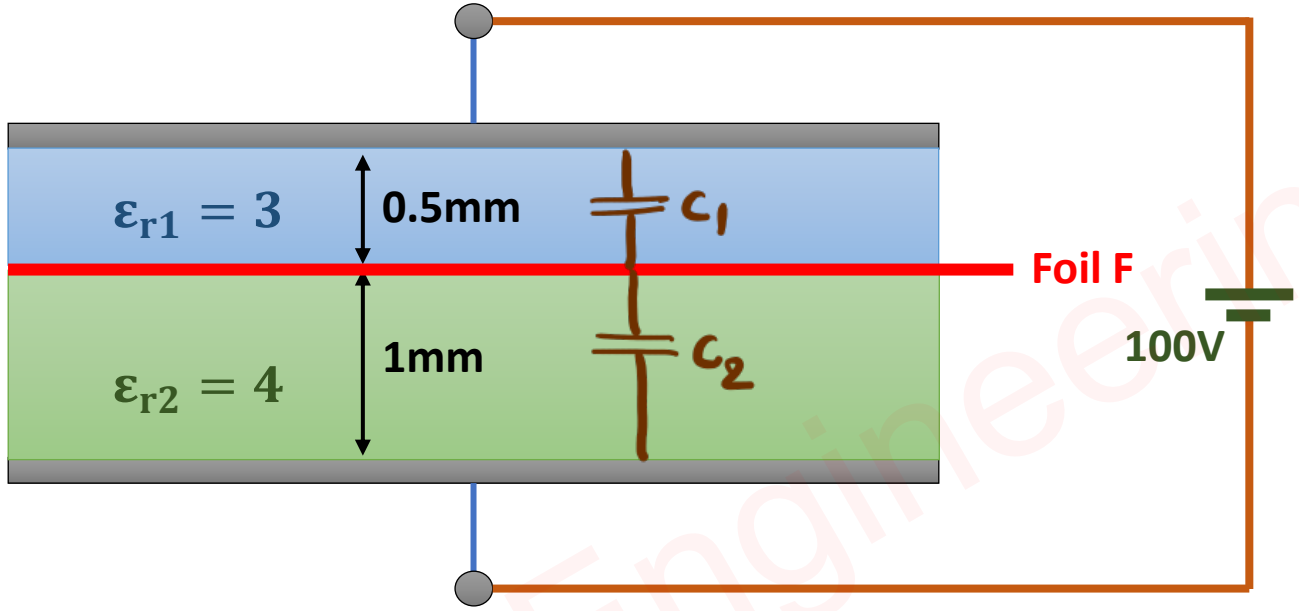
$\rightsquigarrow V_{AB} = V_2 + V_3$

$= \frac{4\mu C}{5\mu F} + \frac{4\mu C}{2\mu F} = 2.8$ Volt

Engineering Funda

Examples of Capacitor & Capacitance

Question 11 – A Composite parallel capacitor is made up of two different materials with different thicknesses as shown in the figure. The two different materials are separated by conducting foil F. The voltage of the conducting foil is ……60 Volt……



$$\rightsquigarrow V_F = V_{C_2} = V\left[\frac{C_1}{C_1+C_2}\right]$$

$$\rightsquigarrow C_1 = \varepsilon_0\varepsilon_{r1}\left(\frac{A}{d_1}\right) = \varepsilon_0(3)\left(\frac{A}{0.5\times10^{-3}}\right)$$

$$= \varepsilon_0 A(6000)$$

$$\rightsquigarrow C_2 = \varepsilon_0\varepsilon_{r2}\left(\frac{A}{d_2}\right) = \varepsilon_0(4)\left(\frac{A}{10^{-3}}\right)$$

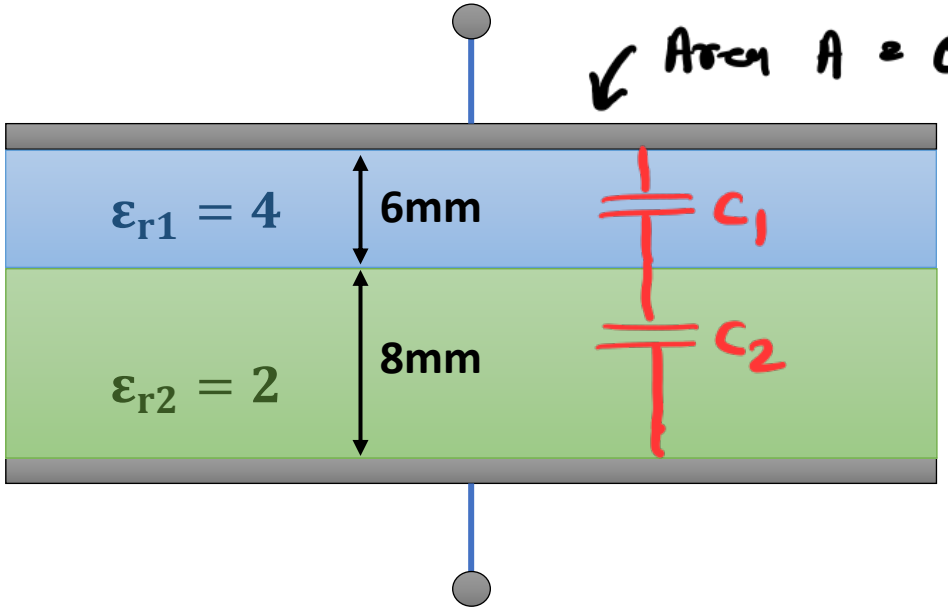
$$= \varepsilon_0 A(4000)$$

$$\rightsquigarrow V_F = 100\left(\frac{\cancel{\varepsilon_0}A(6000)}{\cancel{\varepsilon_0}A(6000)+\cancel{\varepsilon_0}A(4000)}\right)$$

$$= 60 \text{ Volt}$$

Engineering Funda

Question 12 – A parallel capacitor is shown in the figure. It is made of two square metal plates of 400mm side. The 14mm space between the plates is filled with two layers of dielectric material as shown in the figure. The total capacitance is .....



Area $A = 0.4 \times 0.4 = 0.16\,m^2$

$$\rightsquigarrow C_1 = \varepsilon_0 \varepsilon_{r1}\left(\frac{A}{d_1}\right) = 8.854\times10^{-12}\times 4 \times \frac{0.16}{6\times10^{-3}} = 0.944\ nF$$

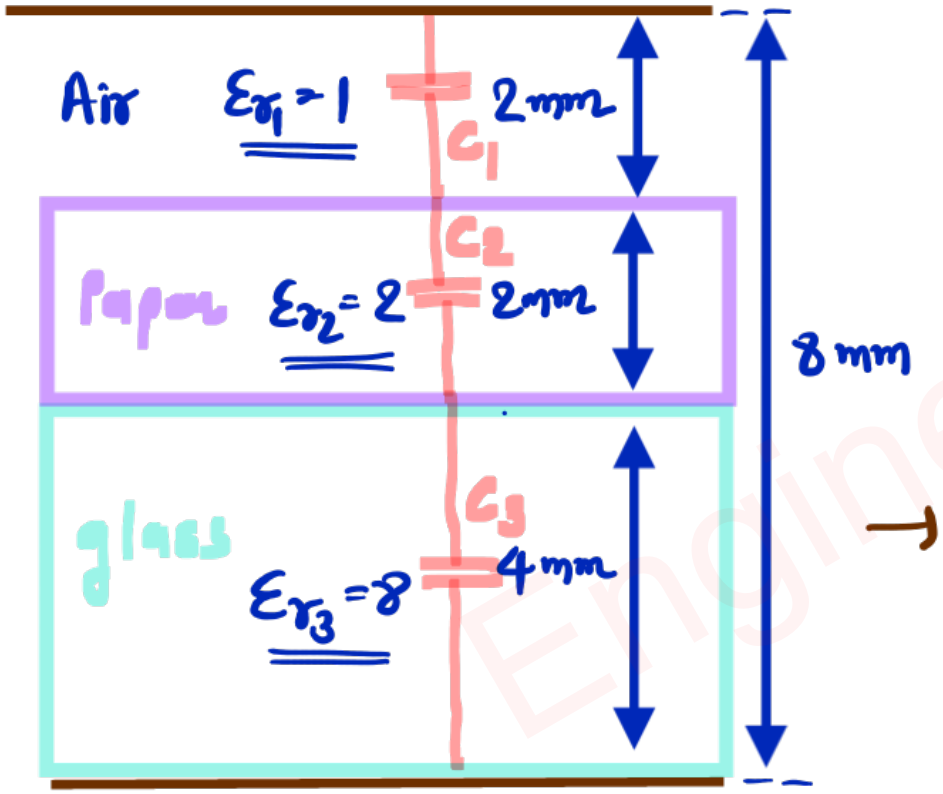
$$\rightsquigarrow C_2 = \varepsilon_0 \varepsilon_{r2}\left(\frac{A}{d_2}\right) = 8.854\times10^{-12}\times 2 \times \frac{0.16}{8\times10^{-3}} = 0.354\ nF$$

$$\rightsquigarrow C = \frac{C_1 C_2}{C_1 + C_2} = \frac{0.944\times0.354}{0.944+0.354} = 0.257\ nF$$

Engineering Funda

Question 13 – A capacitor consists of two metal plates each $500\text{mm} \times 500\text{mm}$ and spaced 8mm apart. The space between the metal plate is filled with glass and paper with thicknesses 4mm and 2mm, respectively. The relative permittivity of glass and paper are 8 and 2, respectively. The capacitance is ....632 pF.... or 0.632 nF

$A = 0.5 \times 0.5 = 0.25\ m^2$



$\rightsquigarrow C_1 = \varepsilon_0 \varepsilon_{r_1} \left(\frac{A}{d_1}\right) = 8.854 \times 10^{-12} \times 1 \times \frac{0.25}{2 \times 10^{-3}} = 1.106\ nF$

$\rightsquigarrow C_2 = \varepsilon_0 \varepsilon_{r_2} \left(\frac{A}{d_2}\right) = 8.854 \times 10^{-12} \times 2 \times \frac{0.25}{2 \times 10^{-3}} = 2.213\ nF$

$\rightsquigarrow C_3 = \varepsilon_0 \varepsilon_{r_3} \left(\frac{A}{d_3}\right) = 8.854 \times 10^{-12} \times 8 \times \frac{0.25}{4 \times 10^{-3}} = 4.427\ nF$

$\rightarrow$ For Series

$\Rightarrow \frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$

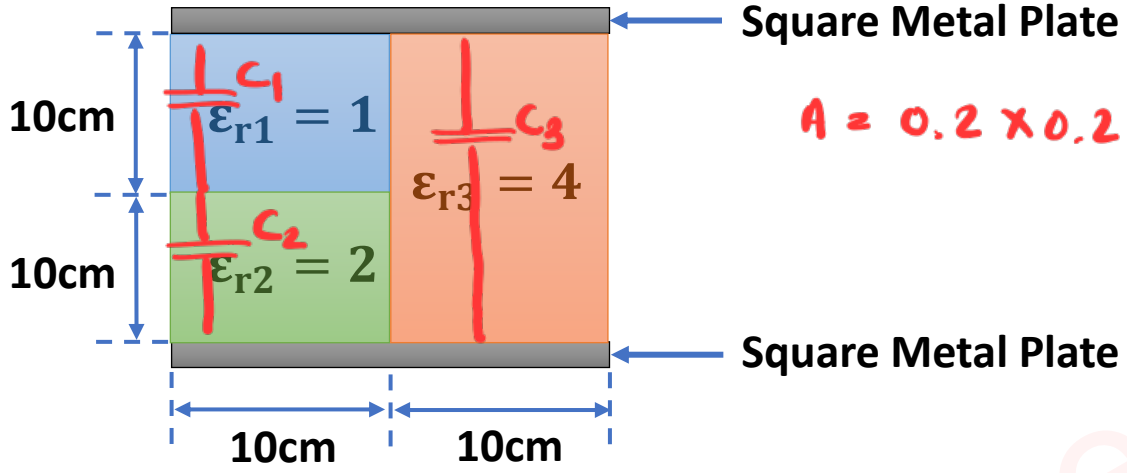
$\Rightarrow \frac{1}{C} = \frac{1}{1.106} + \frac{1}{2.213} + \frac{1}{4.427}$

$\Rightarrow \frac{1}{C} = 1.5819 \Rightarrow C = 0.632\ nF = 632\ pF$

Engineering Funda

Examples of Capacitor & Capacitance

Question 14 – The capacitance of the arrangement shown in the figure is ……4.72…… pF



$A = 0.2 \times 0.2 = 0.04 m^2$

$\rightsquigarrow C_1 = \varepsilon_0 \varepsilon_{r1} \frac{A_1}{d_1} = 8.854 \times 10^{-12} \times 1 \times \frac{0.02}{0.1}$

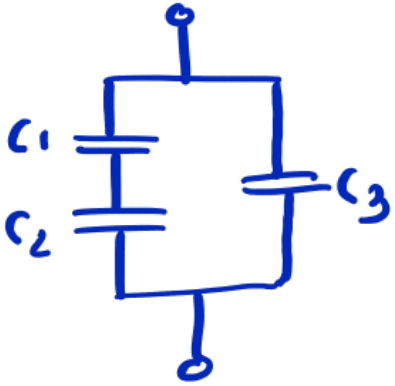
$= 1.77$ pF

$\rightsquigarrow C_2 = \varepsilon_0 \varepsilon_{r2} \frac{A_2}{d_2} = 8.854 \times 10^{-12} \times 2 \times \frac{0.02}{0.1}$

$= 3.54$ pF

$\rightsquigarrow C_3 = \varepsilon_0 \varepsilon_{r3} \frac{A_3}{d_3} = 8.854 \times 10^{-12} \times 4 \times \frac{0.02}{0.2}$

$= 3.54$ pF



$\rightarrow C = C_3 + C_1 \approx C_2$

$= C_3 + \frac{C_1 C_2}{C_1 + C_2}$

$= 3.54 + \frac{1.77 \times 3.54}{1.77 + 3.54}$

$= 4.72$ pF

Engineering Funda

Question 15 – The capacitance of an isolated sphere of radius 10cm in air is equal to 11.12 pF

→ For Spherical Capacitor

$$\Rightarrow C = \frac{4\pi\varepsilon_0\varepsilon_r}{\left(\frac{1}{a} - \frac{1}{b}\right)}$$

$$= \frac{4 \times 3.14 \times 8.854 \times 10^{-12} \times 1}{\left[\frac{1}{0.1} - \frac{1}{\infty}\right]}$$

$$= 11.12 \times 10^{-12}\ F$$

$$= 11.12\ pF$$

Engineering Funda

Question 16 – A parallel plate capacitor has an electrode area $100mm^2$, with spacing 0.1mm between the electrodes. The dielectric between the plates is air. The voltage across capacitor is 100V. The energy stored in capacitor is ....4.427×10^{-8} J

$\rightsquigarrow A = 100\ mm^2 = 10^{-4}\ m^2$

$d = 0.1\ mm = 10^{-4}\ m$

$\varepsilon_r = 1$

$V = 100$

$\rightsquigarrow C = \varepsilon_0 \varepsilon_r \frac{A}{d}$

$= 8.854 \times 10^{-12} \times 1 \times \frac{10^{-4}}{10^{-4}}$

$= 8.854 \times 10^{-12}\ F$

$\rightarrow E = \frac{1}{2} C V^2$

$= \frac{1}{2} \times 8.854 \times 10^{-12} \times 10^{4}$

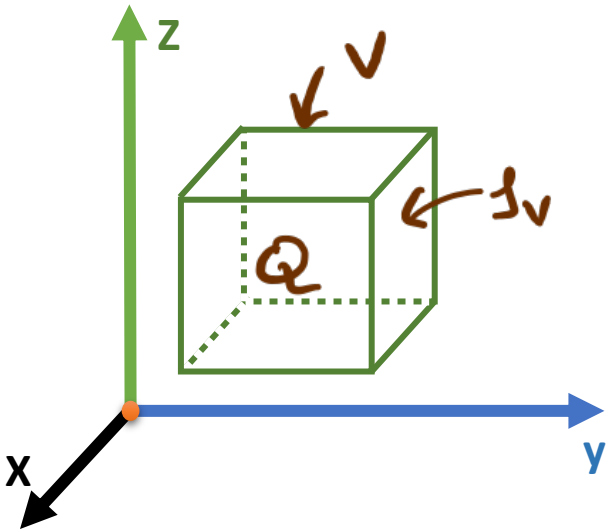
$= 4.427 \times 10^{-8}\ J$

$= 44.27 \times 10^{-9}\ J$

$= 44.27\ nJ$

Engineering Funda

Continuity of Current



~ For closed surface.

$$\Rightarrow I = -\frac{dQ}{dt} \quad \text{—(1)}$$

[∵ (−ve) sign is due to decrease in charge w.r.t time t.]

~ For closed surface

$$\Rightarrow I = \oint \vec{J} \cdot d\vec{s} \quad \text{—(2)}$$

~ From (1) & (2)

$$\Rightarrow \oint \vec{J} \cdot d\vec{s} = -\frac{dQ}{dt} \quad \text{—(A)}$$

~ From Gauss Divergence Theorem.

$$\Rightarrow \oint \vec{J} \cdot d\vec{s} = \int \vec{\nabla} \cdot \vec{J} \, dV \quad \text{—(3)}$$

Engineering Funda

~ From Vol.^{m} charge density ρ_v

$$\Rightarrow Q = \int \rho_v \, dV \quad \text{— (4)}$$

~ Put (3) and (4) in eq.^{n} (A)

$$\Rightarrow \int \vec{\nabla} \cdot \vec{J} \, dV = -\frac{d \int \rho_v dV}{dt}$$

$$\Rightarrow \int \vec{\nabla} \cdot \vec{J} \, dV = -\int \frac{d\rho_v}{dt} \, dV$$

$$\Rightarrow \vec{\nabla} \cdot \vec{J} = -\frac{d\rho_v}{dt} \quad \text{— (B)}$$

~ Eq.^{n} (B) is eq.^{n} of Continuity of Current.

Engineering Funda

Poisson's & Laplace Equation

- Poisson's and Laplace's contribution is enormous in Electromagnetics Field Theory.

Simeon Denis Poisson

Pierre Simon Laplace

Engineering Funda

Derivation of Poisson's Equation

- Electric field based on potential is given by:

$$\Longrightarrow \vec{\mathbf{E}} = -\vec{\nabla}\mathbf{V}$$

- As per Gauss's Law of Electric Field:

$$\Longrightarrow \vec{\nabla}.\vec{\mathbf{E}} = \frac{\rho_V}{\varepsilon_0}$$

- By Applying the Equation of Electric Field in Gauss' Law of Electric Field:

$$\Longrightarrow \vec{\nabla}.\left(-\vec{\nabla}\mathbf{V}\right) = \frac{\rho_V}{\varepsilon_0}$$

$$\Longrightarrow \vec{\nabla}.\left(\vec{\nabla}\mathbf{V}\right) = -\frac{\rho_V}{\varepsilon_0}$$

$$\Longrightarrow \nabla^2\mathbf{V} = -\frac{\rho_V}{\varepsilon_0}$$

$[\therefore \nabla^2$ is known as Laplacian Operator$]$

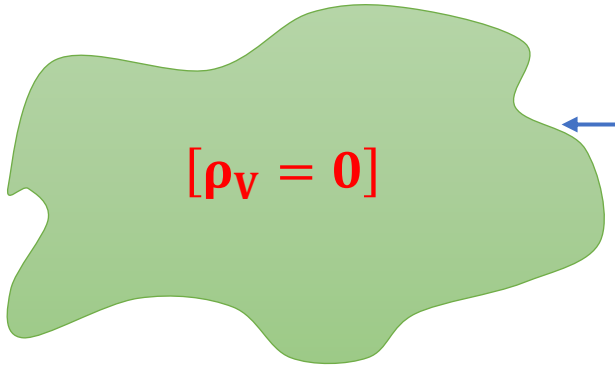
- The Above equation is Poisson's equation.

Engineering Funda

Derivation of Laplace Equation

- Poisson's Equation is given by:

$$\Longrightarrow \nabla^2 V = -\frac{\rho_V}{\varepsilon_0}$$



If the Enclosed surface doesn't have any volume charge ρ_V

- Hence, Poisson's Equation is translated into:

$$\Longrightarrow \nabla^2 V = 0$$

- The Above equation is a Laplace equation.

Find Voltage V Using Passion's & Laplace Equation. → Find E Field $\Longrightarrow \vec{E} = -\vec{\nabla} V$ → Find Capacitance $\Longrightarrow C = \frac{Q}{V}$

Engineering Funda

Applications of Poisson's and Laplace Equation

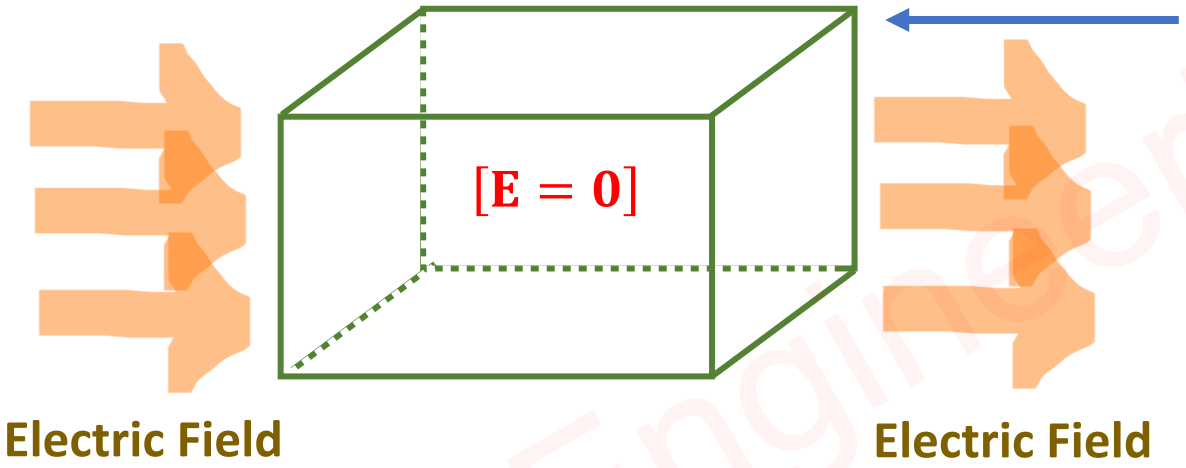
- Poisson's Equation is given by:

$$\Rightarrow \nabla^2 V = -\frac{\rho_V}{\varepsilon_0}$$

- Laplace Equation is given by:

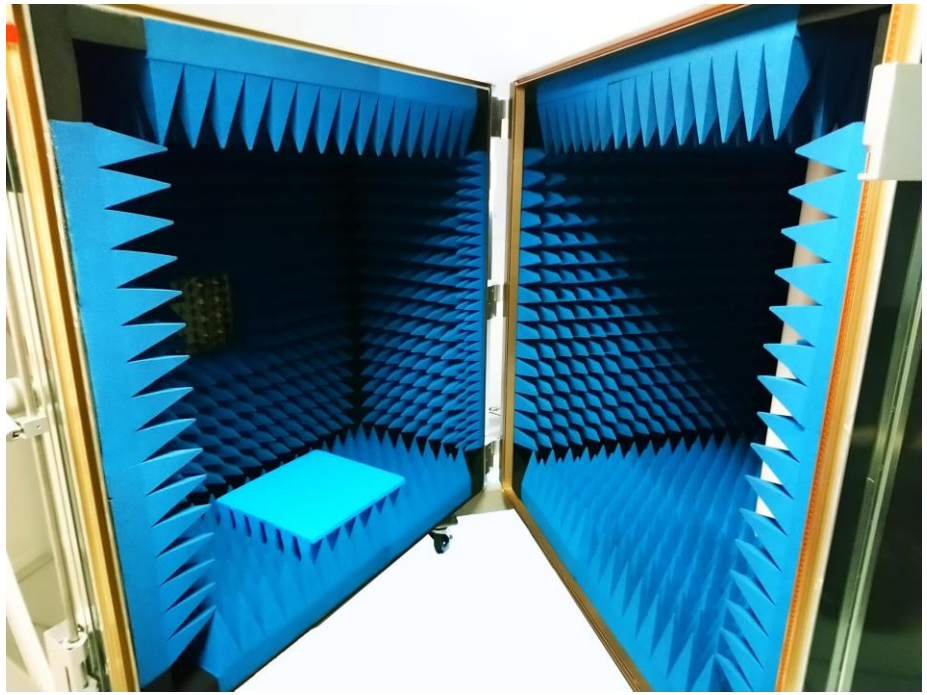
$$\Rightarrow \nabla^2 V = 0$$

- Poisson's equation is crucial when there are sources (like charge or heat), while Laplace's equation applies in source-free regions.



Metal Box is Grounded. Hence, the Voltage on the Box is 0.

- Metallic Grounding is useful in Anechoic Chambers.



Engineering Funda

Uniqueness Theorem

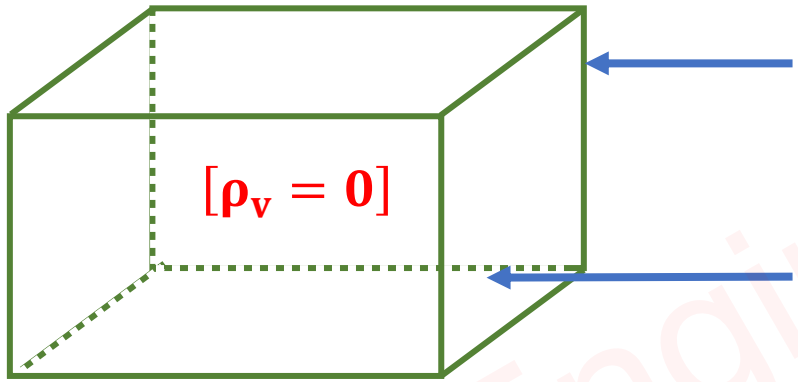
- Poisson's Equation is given by:

$$\Rightarrow \nabla^2 V = -\frac{\rho_V}{\varepsilon_0}$$

- Laplace Equation is given by:

$$\Rightarrow \nabla^2 V = 0$$

- Poisson's and Laplace's equations are applied to the enclosed surface.
- The Uniqueness Theorem explains that there is only one solution to $\nabla^2 V = 0$.



Metal Box will have constant voltage on the surface if the metal box encloses no charge.

Inside Metal Box, There is no Maxima and Minima, If Metal Box doesn't carry any charge.

- The Uniqueness Theorem is applicable to Metallic and Non-Metallic enclosed bodies. With Metallic Bodies, It can be applied on the surface and inside the surface. With Non-Metallic Bodies, it can be applied inside the surface only.

Engineering Funda

Proof of Uniqueness Theorem

- Let's Assume, There are two solutions V_1 and V_2 on the surface.

$$\Rightarrow \nabla^2 V_1 = 0 \qquad \Rightarrow \nabla^2 V_2 = 0$$

- Let's consider, $V = V_1 - V_2$

$$\Rightarrow \nabla^2 V = \nabla^2 V_1 - \nabla^2 V_2 = 0$$

- Hence, $V = 0$, on the surface. It also explains no maxima and minima within given volume.

$$\Rightarrow V = V_1 - V_2 = 0$$

$$\Rightarrow V_1 = V_2$$

- Hence, V_1 and V_2 solutions are same, means there is only one solution of $\nabla^2 V = 0$.

Engineering Funda

Image Theory of Charges

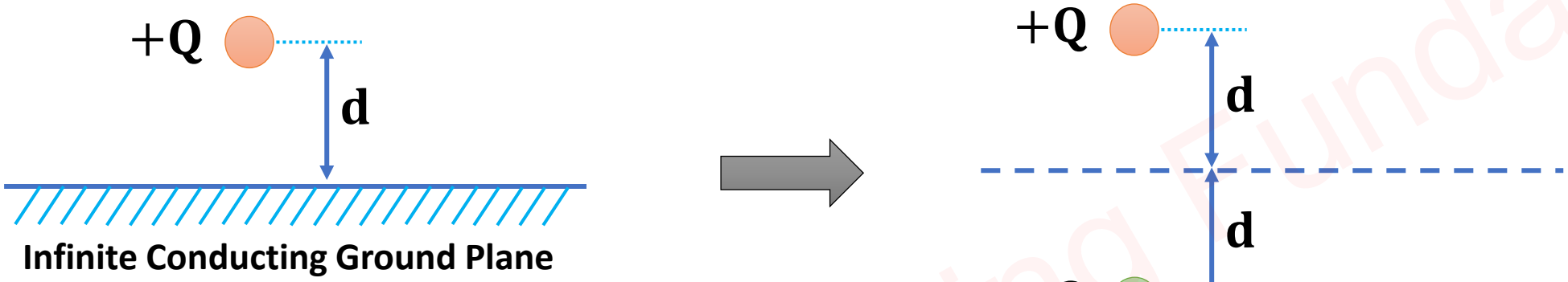
Outlines

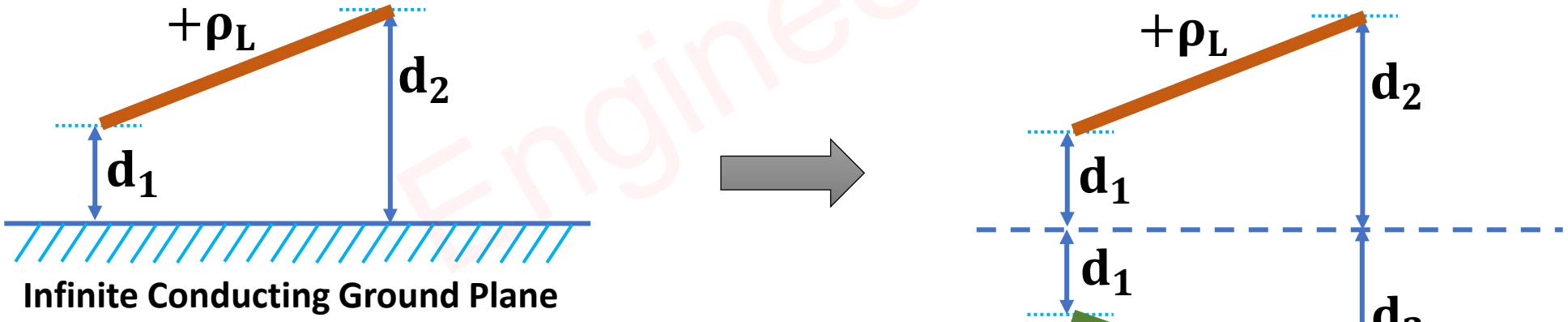
- Basics of image theory of charges
- Examples of image charges
- Important points of image theory
- Image Theory for two ground planes

Engineering Funda

Basics of Image Theory of Charges

- It states that if any charge is placed above Infinite Conducting Ground Plane then Ground Plane is replaced by image of the charge with opposite polarity.





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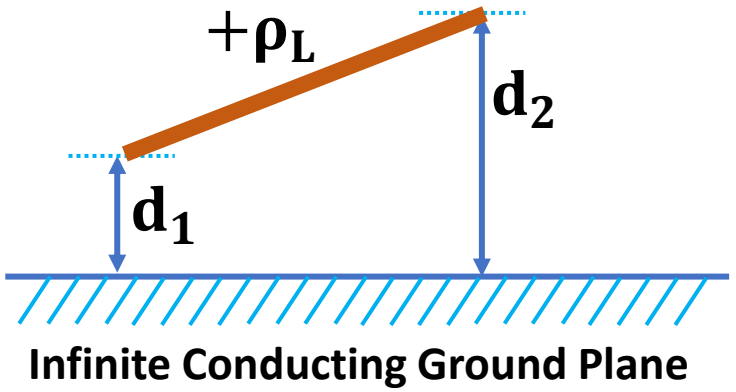
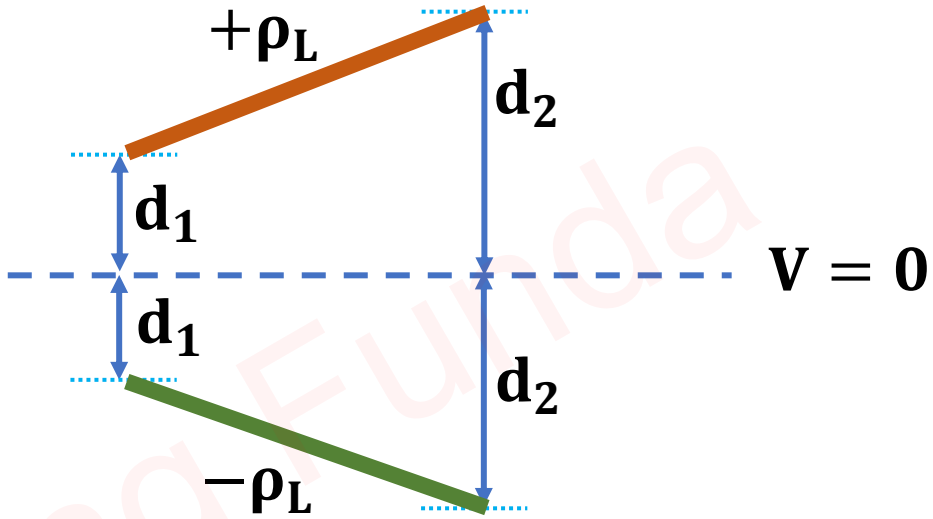
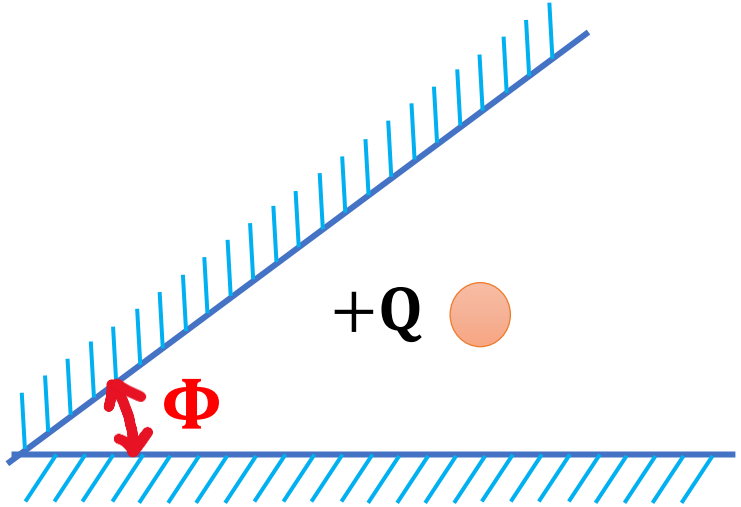
Key Points of Image Theory of Charges





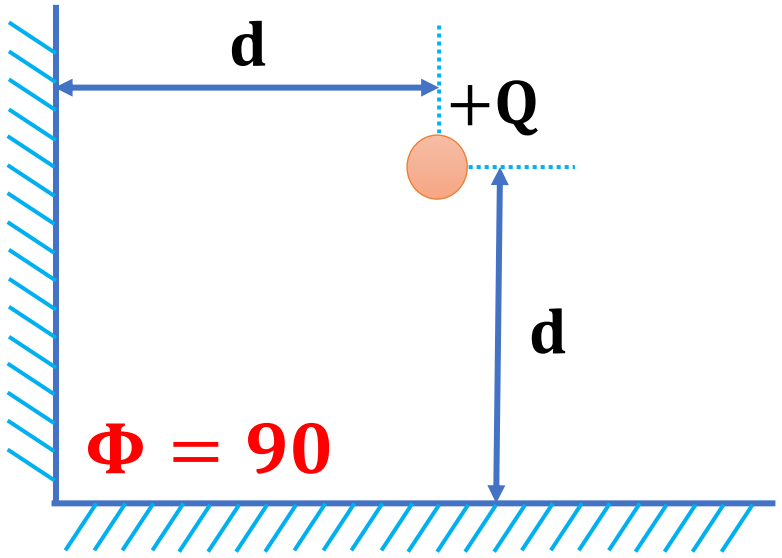
- Image Charges must be there inside the conducting ground plane region.
- Image Charges must be located in such a way that on perfect conducting plane, voltage must be zero.
- Number of Image charges $N = \frac{360^0}{\Phi} - 1$

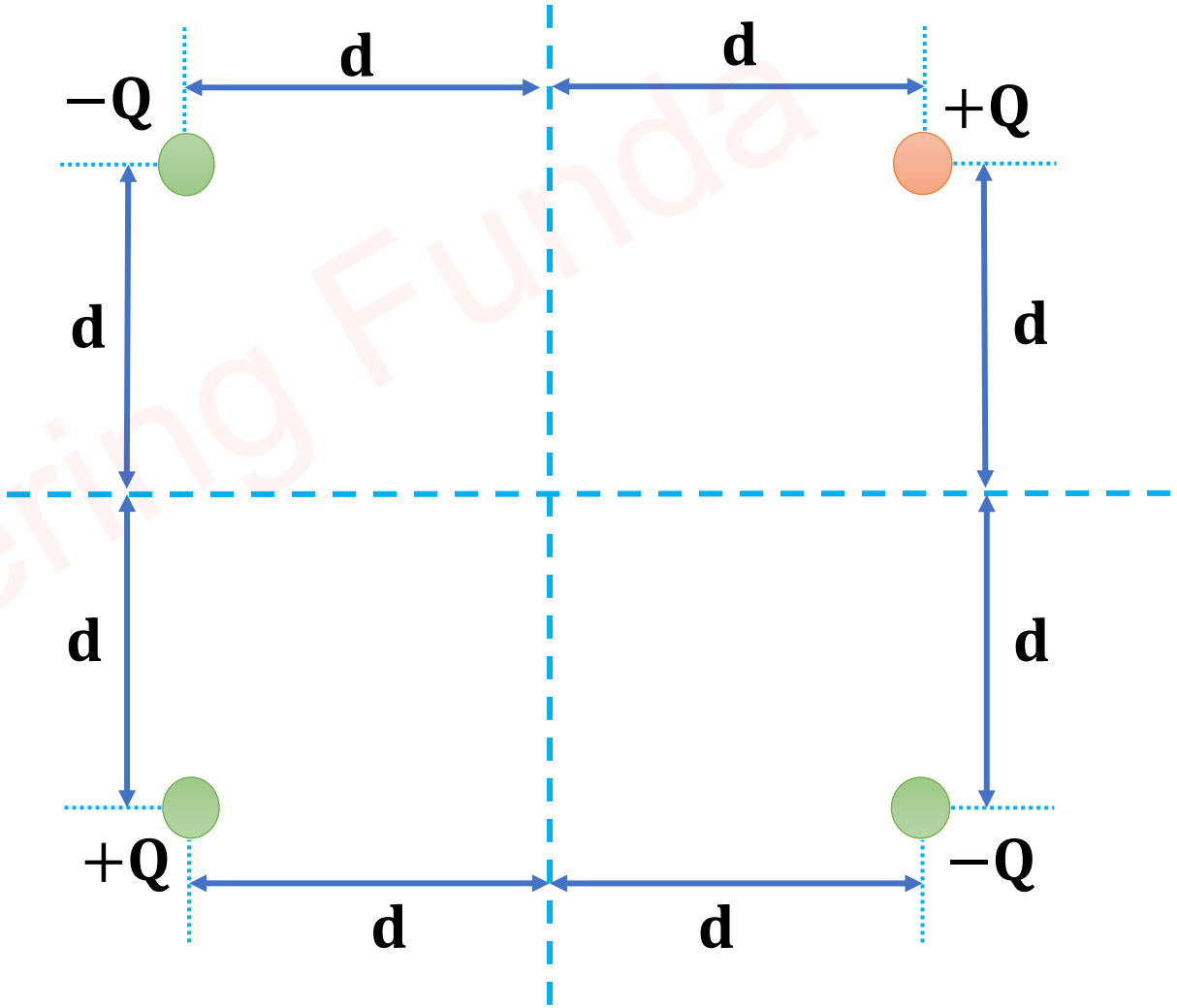


Engineering Funda

Example of Image Theory of Charges

- Find the image charges due to conducting ground plane.





- Number of Image charges

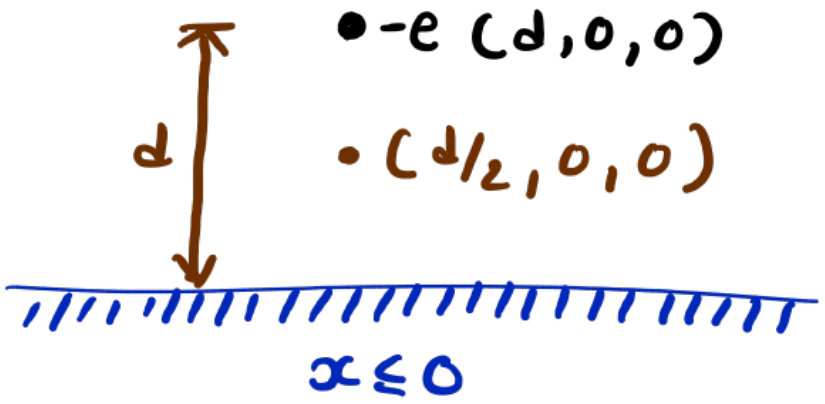
$$\Rightarrow N = \frac{360^0}{\Phi} - 1$$

$$\Rightarrow N = \frac{360^0}{90^0} - 1 = 3$$

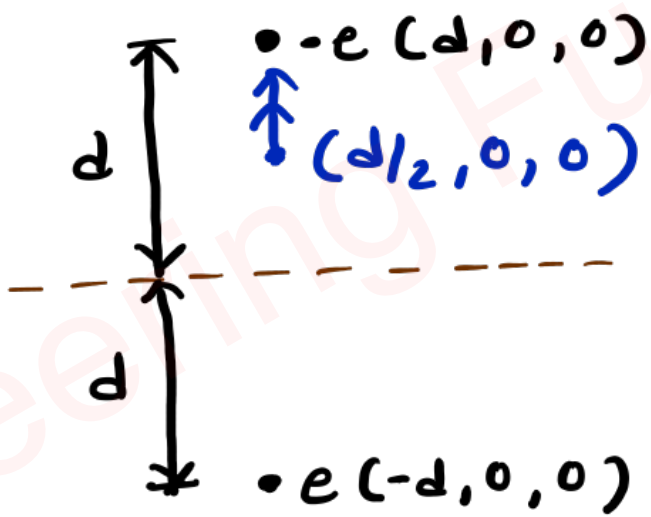
Engineering Funda

Examples of Image Charge

Question 1 – A charge –e is placed in vacuum at the point $(d, 0, 0)$, where $d > 0$. The region $x \leq 0$ is filled uniformly with a metal. The Electric Field at the point $(d/2, 0, 0)$ is .....



$\Rightarrow$



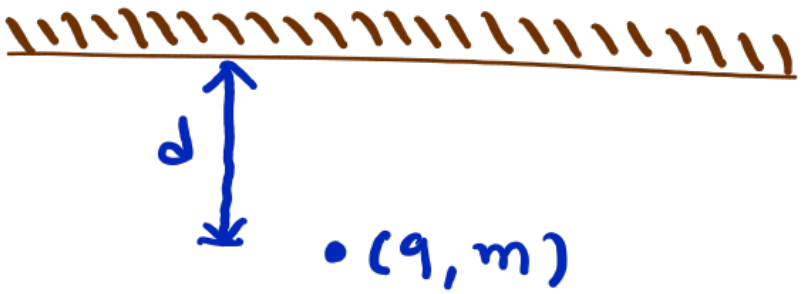
$$\rightarrow E = E_{-e} + E_{+e}$$

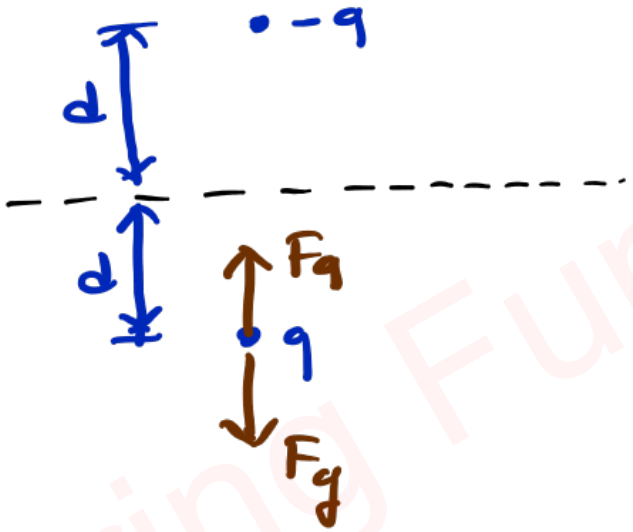
$$= k\frac{e}{(d/2)^2} + k\frac{e}{(3d/2)^2}$$

$$= k\frac{e}{d^2}\left(4 + \frac{4}{9}\right) = \frac{1}{\cancel{4\pi\varepsilon_0}}\frac{e}{d^2}\left(\frac{\overset{10}{\cancel{40}}}{9}\right) = \frac{10e}{9\pi\varepsilon_0 d^2}(1,0,0)$$

Engineering Funda

Question 2 – A point charge q of mass m is kept at a distance d below a grounded infinite conducting sheet in the XY plane. For what value of d will the charge remain stationary?





$$\Rightarrow F_q = F_g$$

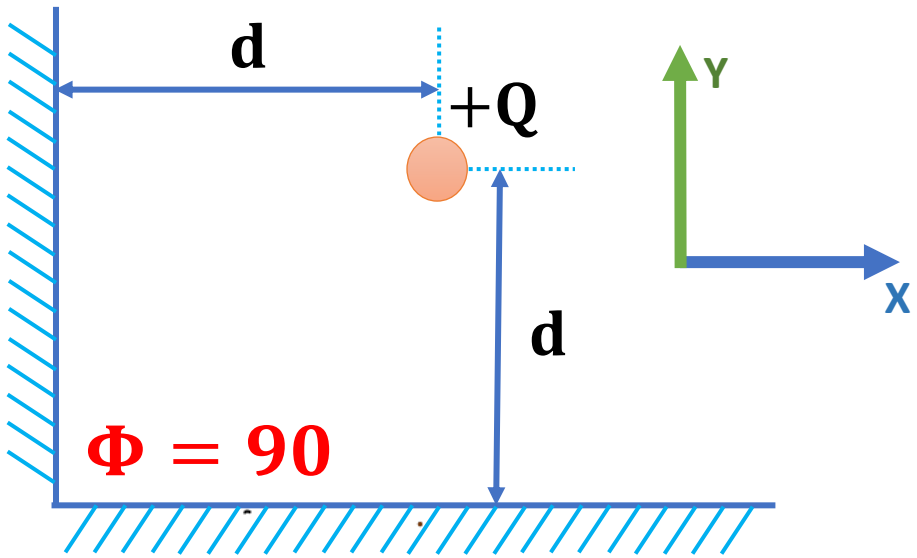
$$\Rightarrow mg = \frac{1}{4\pi\varepsilon_0}\frac{q^2}{(2d)^2}$$

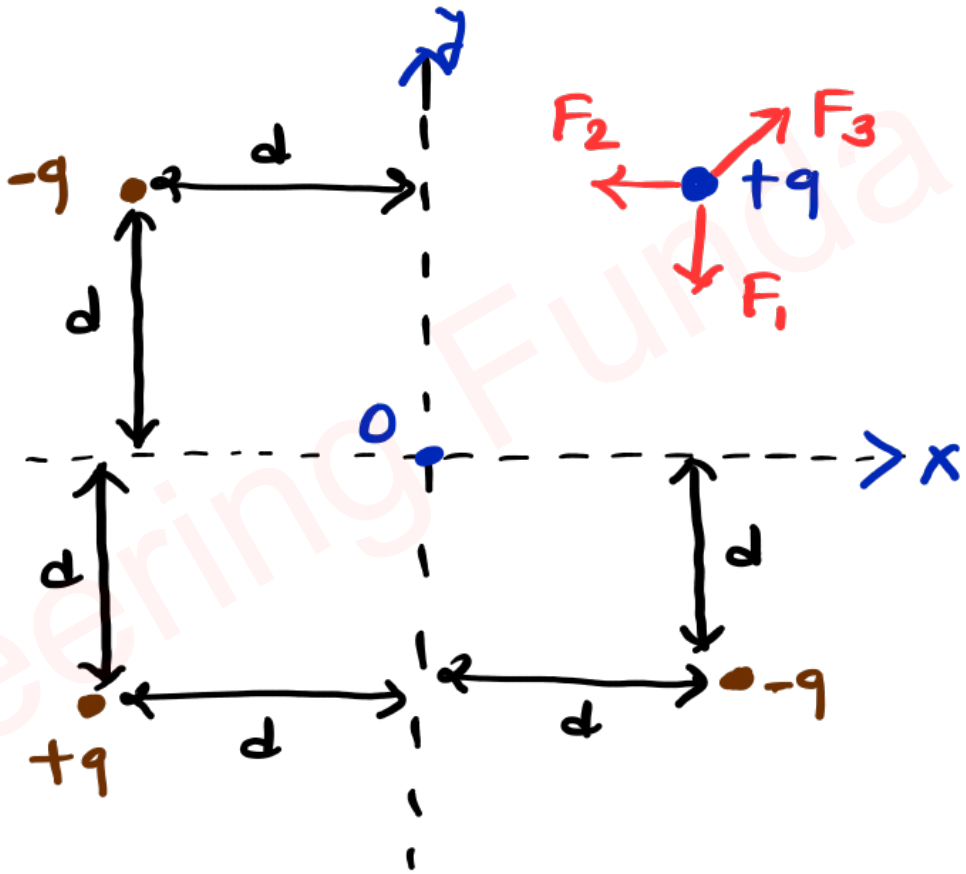
$$\Rightarrow d^2 = \frac{q^2}{16\pi\varepsilon_0 mg}$$

$$\Rightarrow \boxed{d = \frac{q}{4\sqrt{\pi\varepsilon_0 mg}}}$$

Engineering Funda

Question 3 – Net Force on charge Q is given by .....





$$\rightarrow N = \frac{360^\circ}{\phi} - 1$$

$$= \frac{360^\circ}{90^\circ} - 1$$

$$= 3$$

$$\vec{F_1} = k\frac{q^2}{(2d)^2}(-\hat{i}_y), \quad \vec{F_2} = k\frac{q^2}{(2d)^2}(-\hat{i}_x)$$

$$\vec{F_3} = k\frac{q^2}{(2\sqrt{2}d)^2}\left(\frac{\hat{i}_x}{\sqrt{2}} + \frac{\hat{i}_y}{\sqrt{2}}\right)$$

Engineering Funda

$$\rightarrow \vec{F} = \vec{F_1} + \vec{F_2} + \vec{F_3}$$

$$= k\frac{q^2}{d^2}\left[\hat{i}_x\left(-\frac{1}{4}+\frac{1}{8\sqrt{2}}\right) + \hat{i}_y\left(-\frac{1}{4}+\frac{1}{8\sqrt{2}}\right)\right]$$

$$= k\frac{q^2}{d^2}\left[\left(\frac{1-2\sqrt{2}}{8\sqrt{2}}\right)\hat{i}_x + \left(\frac{1-2\sqrt{2}}{8\sqrt{2}}\right)\hat{i}_y\right]$$

$$= k\frac{q^2}{d^2}\left(\frac{1-2\sqrt{2}}{8\sqrt{2}}\right)(\hat{i}_x + \hat{i}_y)$$

$$\rightarrow |\vec{F}| = F = k\frac{q^2}{d^2}\left(\frac{1-2\sqrt{2}}{8\cancel{\sqrt{2}}}\right)\left(\cancel{\sqrt{1+1}}\right)$$

$$= k\frac{q^2}{d^2}\left(\frac{1-2\sqrt{2}}{8}\right)$$

Engineering Funda

Example of Electrostatic Field

Question 1 – In Cylindrical Coordinate System, The potential produced by a uniform ring charge is given by $\Phi(r, z)$, where Φ is a continuous function of r and z. Let $\vec{E}$ be the resulting Electric Field. Then the magnitude of $\nabla \times \vec{E}$ is ……0……

a. Increase with r
b. 0 ✓
c. 3
d. Decrease with Z

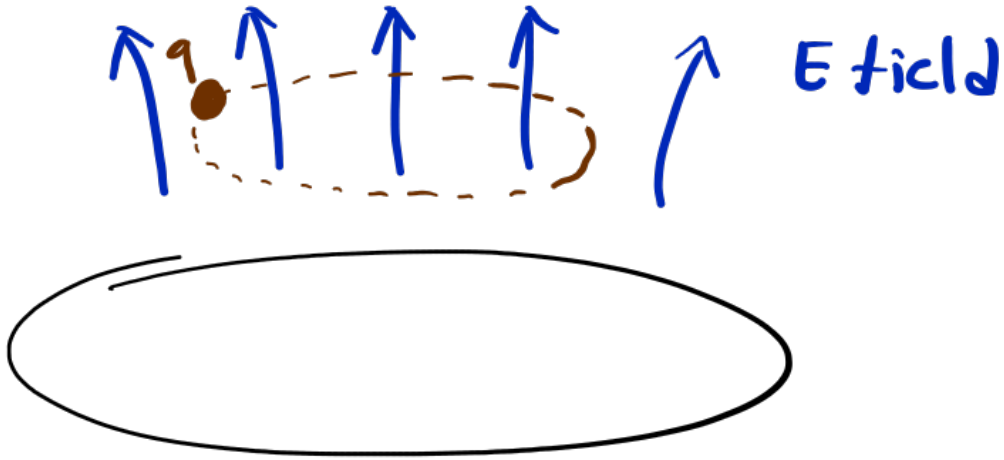
⇝ Here $\vec{E}$ field will not change w.r.t time t.

1) E field is Conservative

2) E field is path Independent

3) E field is Curl Free.

Engineering Funda



$\rightarrow \oint \vec{E} \cdot d\vec{l} = 0$

$\rightarrow$ As per Stocks theorem

$\Rightarrow \oint \vec{E} \cdot d\vec{l} = \int (\vec{\nabla} \times \vec{E}) \cdot d\vec{S} = 0$

$\Rightarrow \boxed{\vec{\nabla} \times \vec{E} = 0}$

Engineering Funda